

Examine how the values and needs of society influence the focus of scientific research

Learning intention

- examine how the values and needs of society influence the focus of scientific research.

Teach and model

- examining why many manufacturers are adopting green chemistry processes.
- exploring how governments determine which scientific research projects should be funded.
- considering innovative energy transfer devices, including those used in transport and communication.

Check understanding

- Explain the main idea and give one accurate example.
- Show the idea in a second way: with words, a model, a diagram or symbols.
- Apply the idea to a short unfamiliar situation and justify the result.

Teaching cautions

- Ask the student to explain their choice, not only state an answer.
- Check that key vocabulary is understood in a new example.
- Mix representations so the skill is transferred rather than copied.

Quick lesson sequence

- Connect to prior knowledge and introduce the key vocabulary.
- Model one clear example, then solve a second example together.
- Finish with an independent check and a short student explanation.